

A sextic counterexample to Euler's sum of powers conjecture

OPG-508

L1 - TRACTABLE

OPEN

Number Theory

ORIGINAL PROBLEM STATEMENT

Problem Find six positive integers $x_1, x_2, \dots, x_6$ such that

$$x_1^6 + x_2^6 + x_3^6 + x_4^6 + x_5^6 = x_6^6$$

or prove that such integers do not exist.